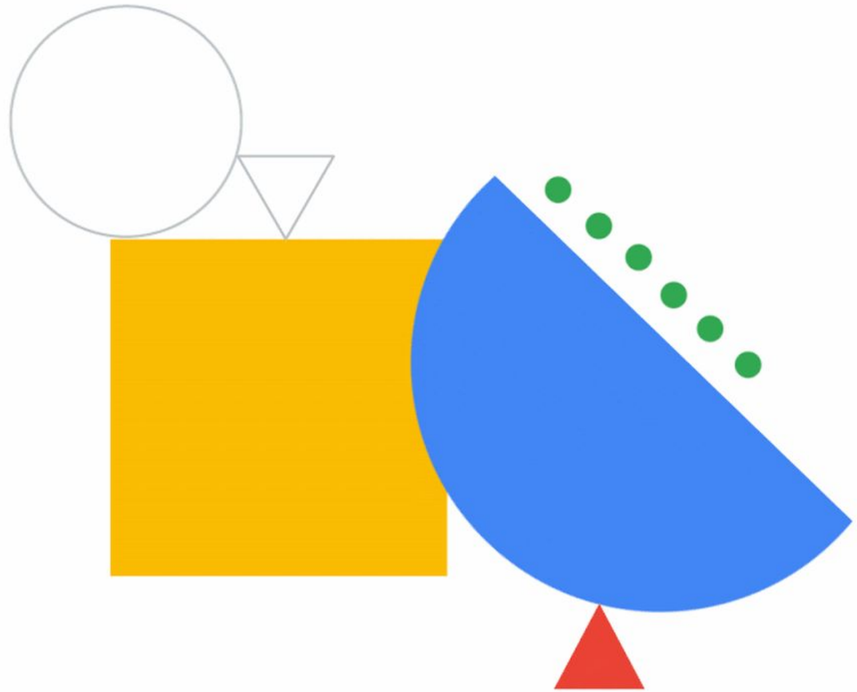
Google Cloud

Advanced Logging and Analysis



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Objectives

01 Use Log Explorer features

02 Explain the features and benefits of log-based metrics

03 Define log sinks (inclusion filters and exclusion filters)

04 Explain how BigQuery can be used to analyze logs

05 Use Log Analytics on Google Cloud



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In this section, you'll explore



- Cloud Logging overview and architecture
- Log types and collection
- Storing, routing, and exporting the logs
- Query and view logs
- Using log-based metrics
- Log Analytics

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Cloud Logging help you understand your application



Gather data from various workloads

Gathers the information required to troubleshoot and understand the workload and application needs



Analyze large volumes of data

Tools like Error Reporting, Log Explorer, and Log Analytics let you drive insights from large sets of data



Route and store logs

Route your logs to the region or service of your choice for additional compliance or business benefits

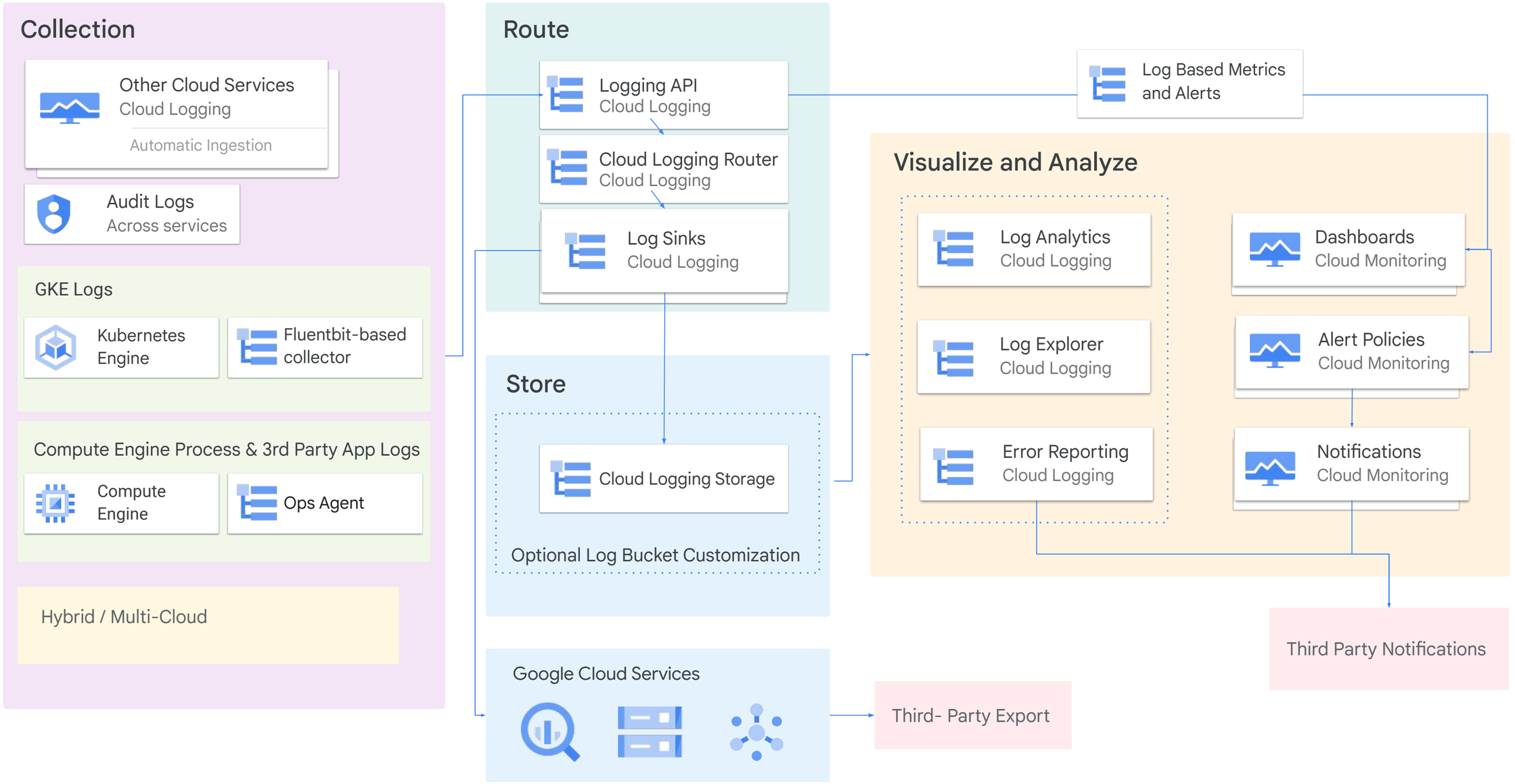


Get compliance insights

Leverage audit and app logs for compliance patterns and issues

Cloud Logging architecture

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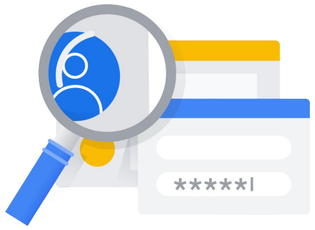
Available logs



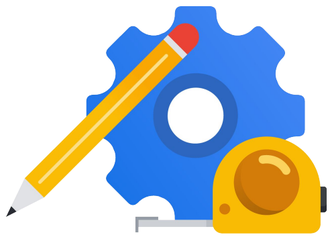
Platform logs



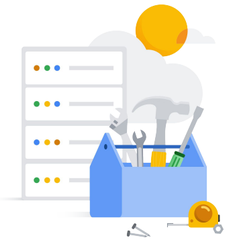
Component logs



Security logs



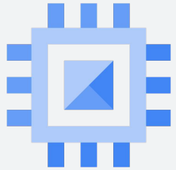
User-written logs



Multi / Hybrid Cloud logs

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Some logs from Google Cloud resources are collected automatically

	Google Kubernetes Engine	Compute Engine	Serverless compute services
			
Cloud Logging	Logs written to **stdout and stderr** are collected automatically	Install the **Ops Agent** on your VMs	Logs written to **stdout** and **stderr** are collected automatically

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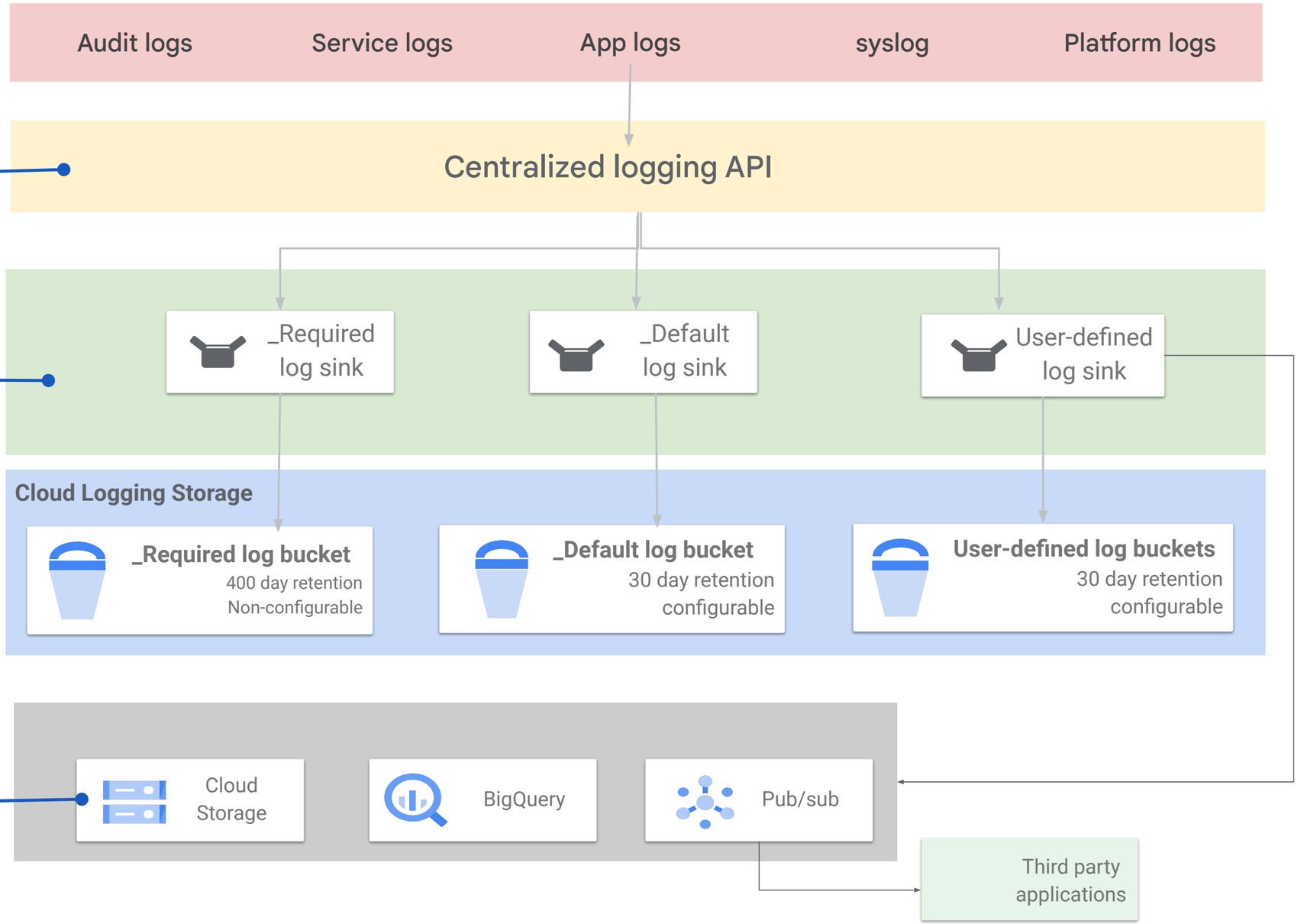
Log routing and storage

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Cloud Logging receives log entries through Cloud Logging API

Sinks contain the inclusion and exclusion filters that determine the log destination

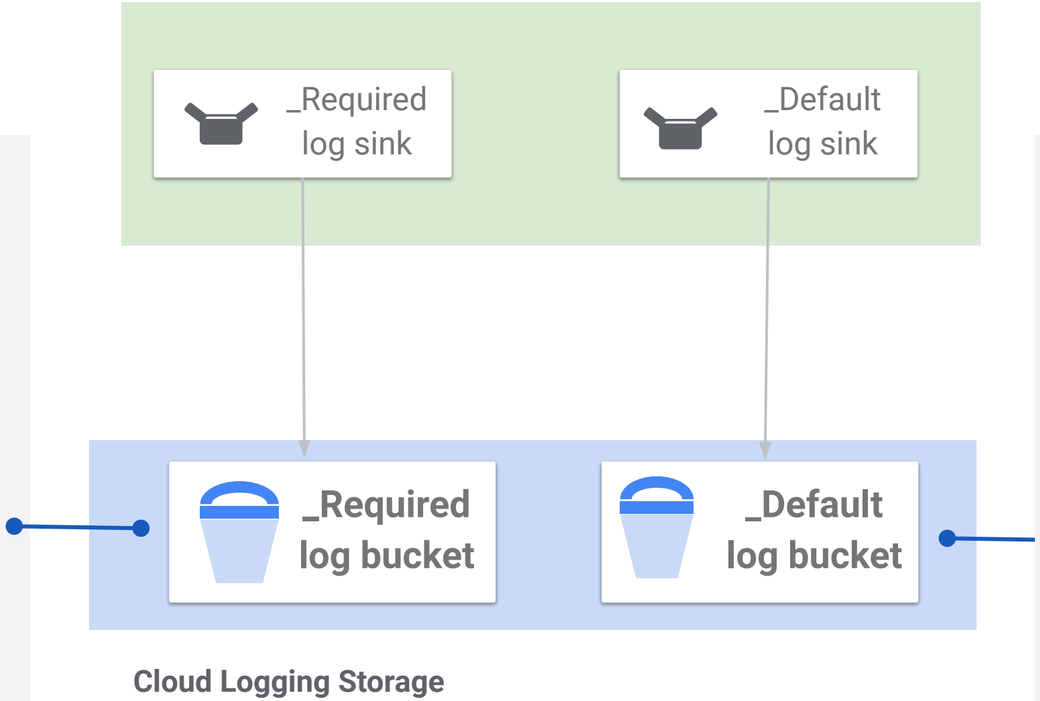
Cloud Storage buckets are different storage entities than Cloud Logging buckets



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_Required

- Routes admin activity, system event and access transparency logs automatically
- No charges
- Retention period: Non-configurable 400 days
- Cannot be deleted or modified

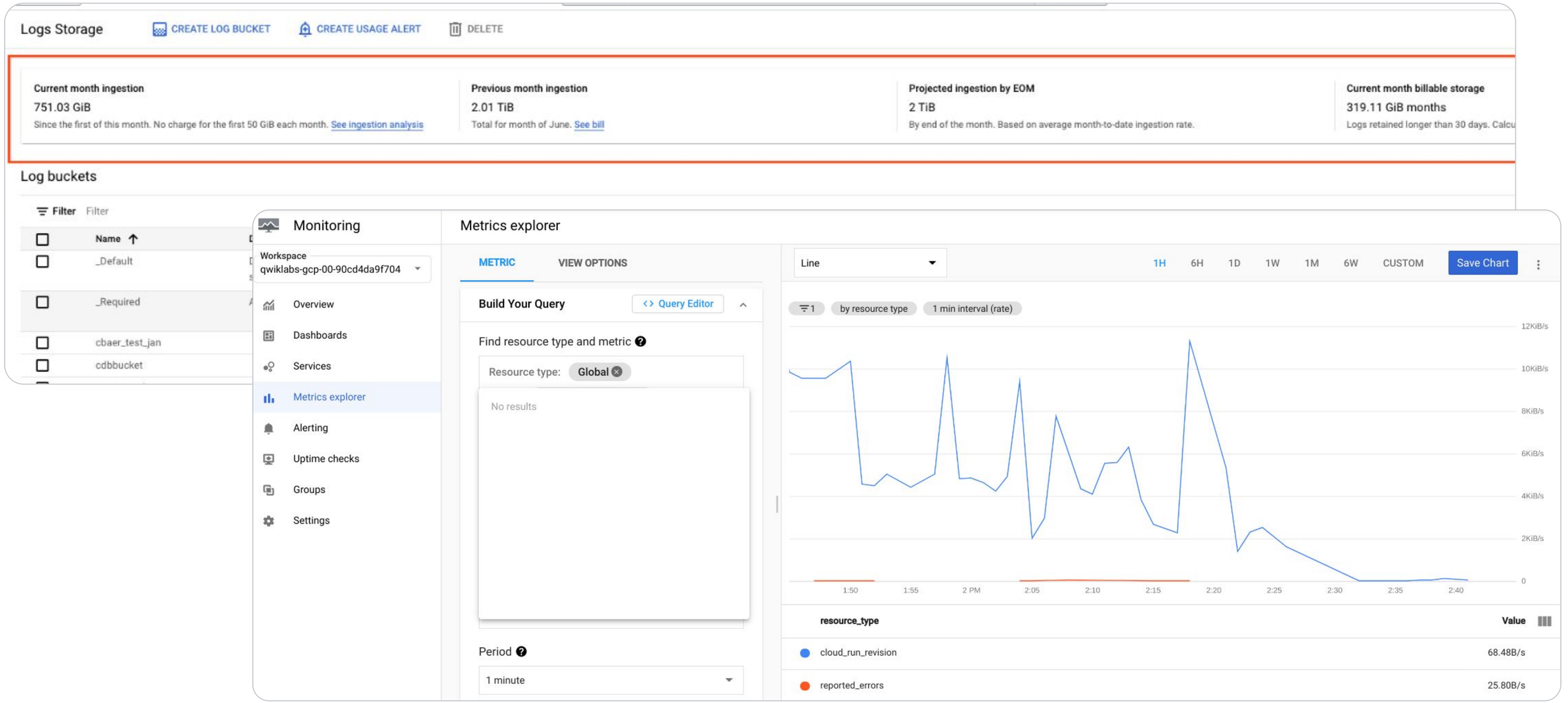


_Default

- Logs not ingested by _required bucket are routed by the _Default sink
- No charges
- Configurable retention period
- Cannot be deleted but can be disabled.

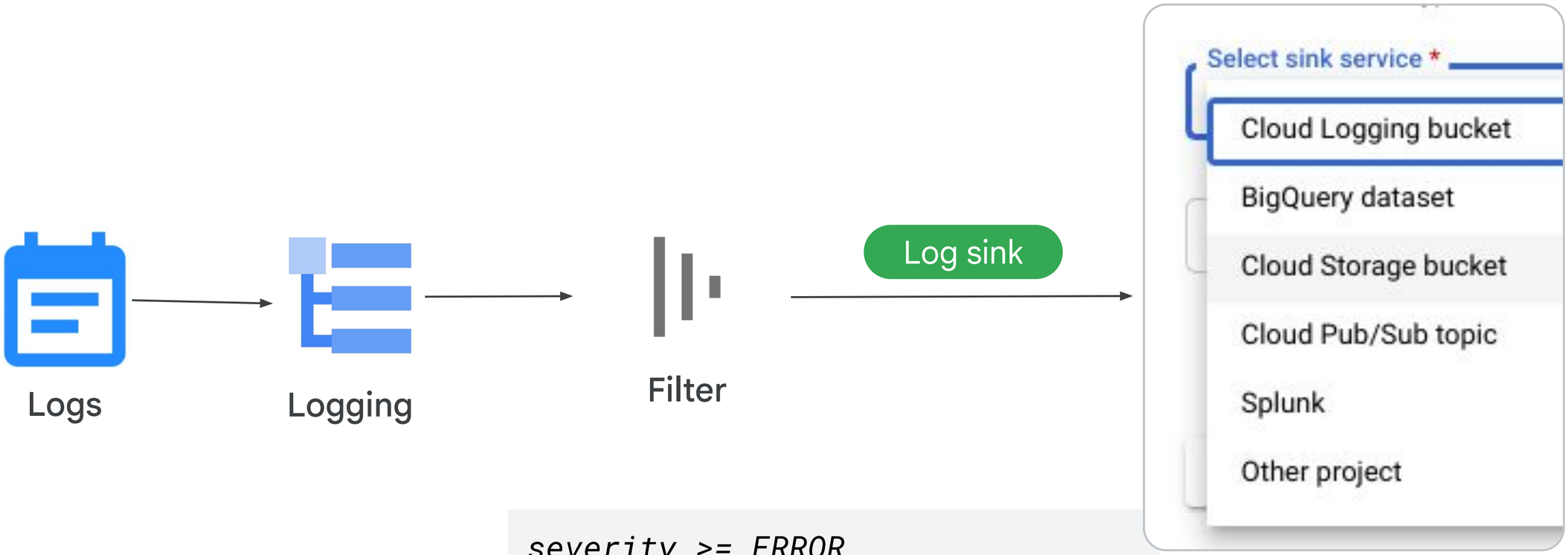
Logs Storage and usage

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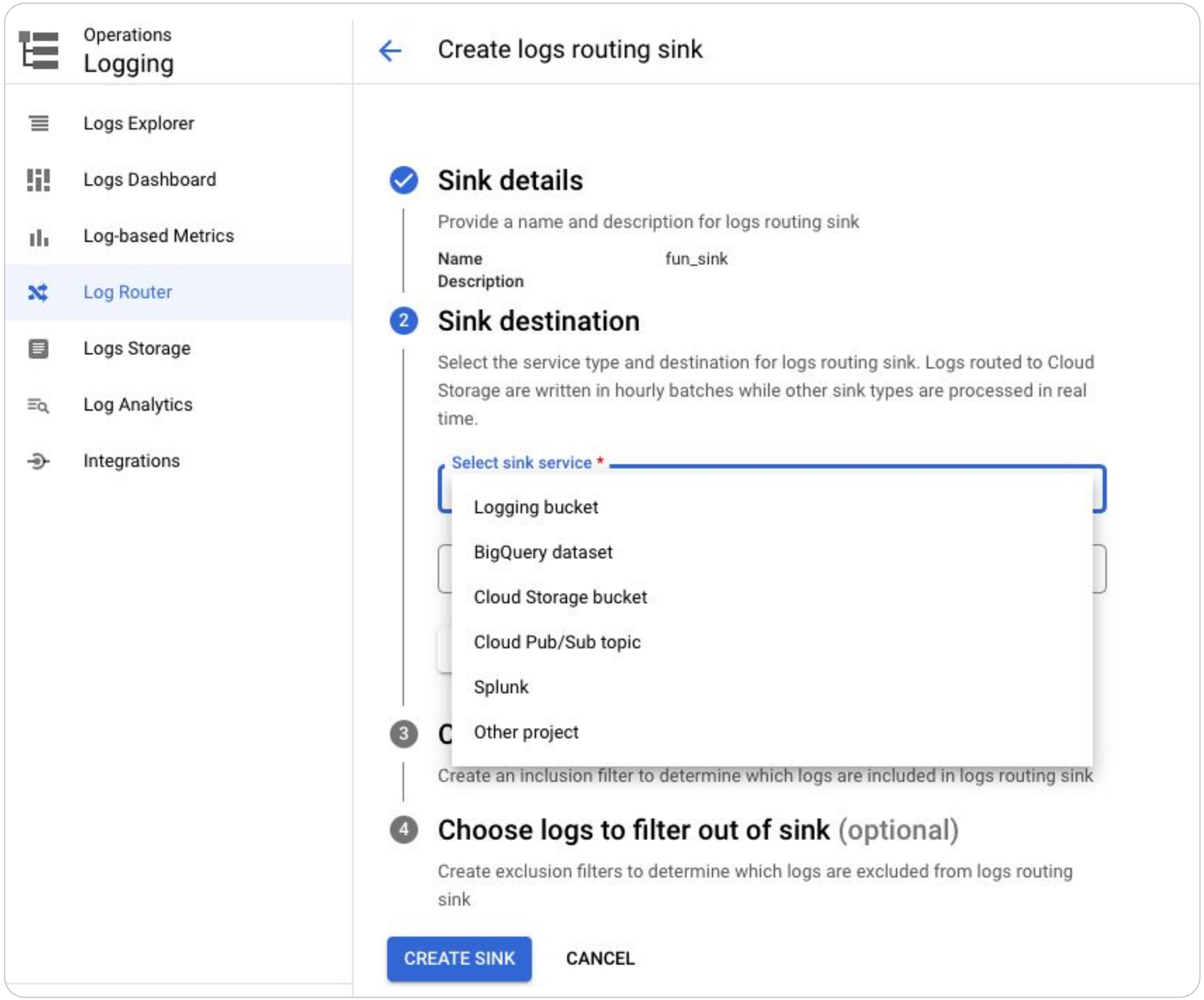
Log router sinks and sink locations




```
severity >= ERROR
resource.type="k8s_container"
resource.labels.project_id="XX"
```

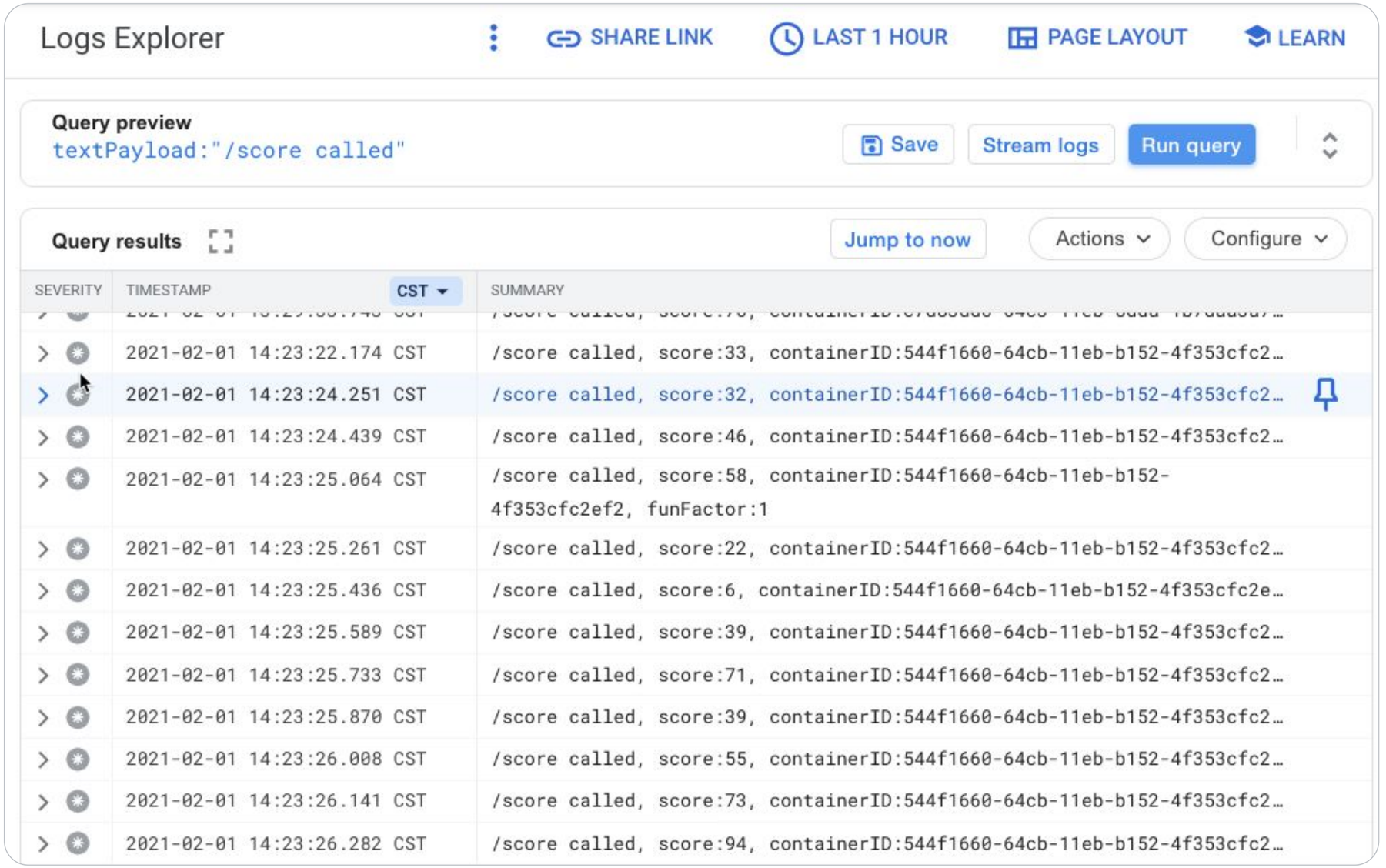
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Create a log sink



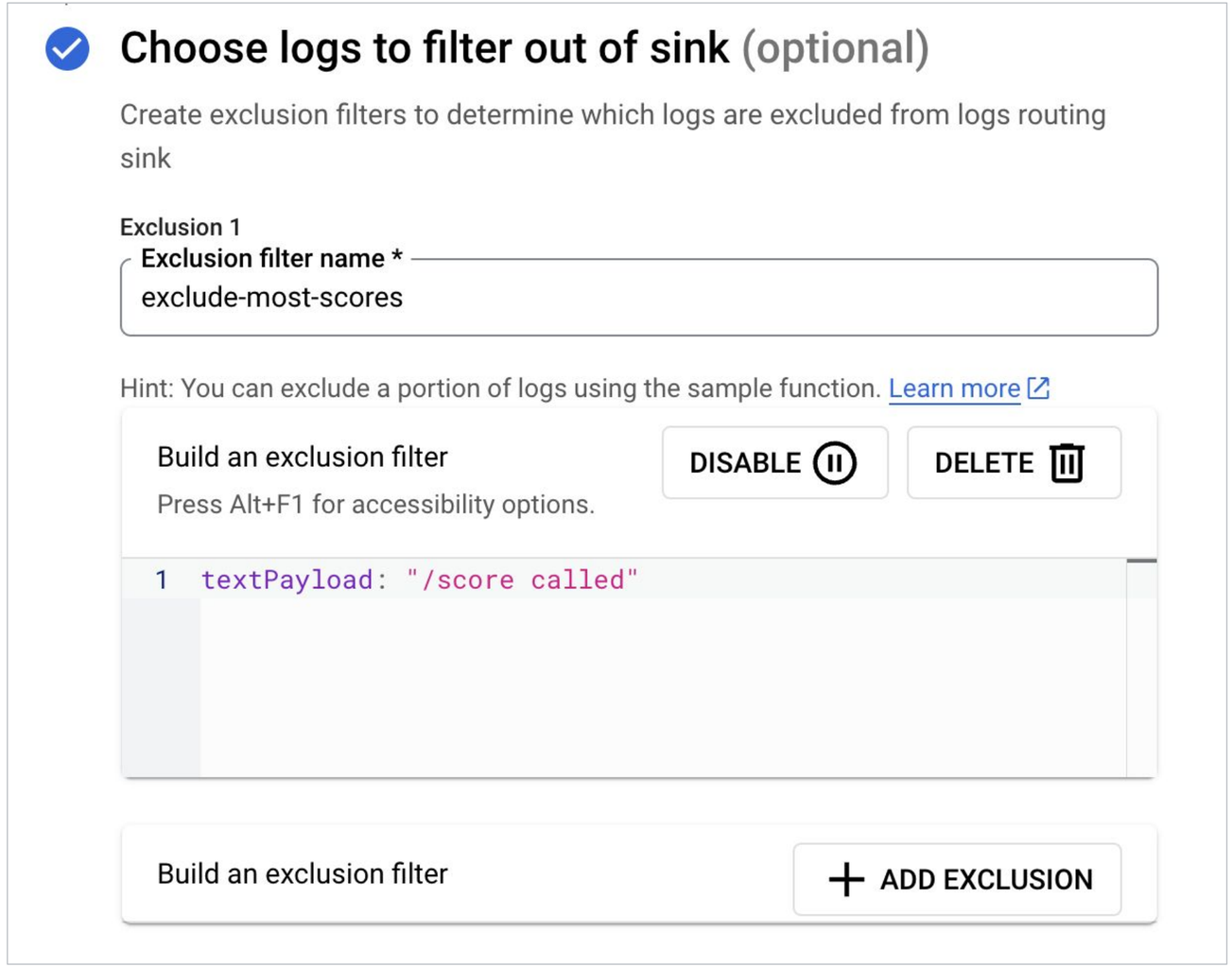
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Exclusions: Identify log entries



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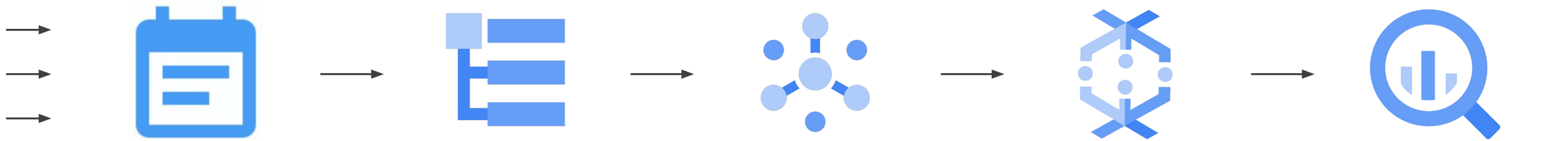
Exclusions: Build the exclusion



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Log archiving and analysis

Example pipeline



1 Events

2 Logging

3 Pub/Sub

4 Dataflow

5 BigQuery

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Archive logs for long-term storage

Example pipeline



1 Events

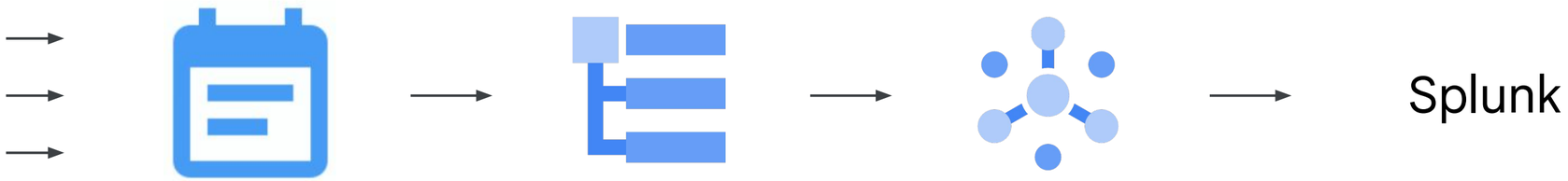
2 Logging

3 Cloud Storage

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Exporting back to Splunk

Example pipeline



1 Events

2 Logging

3 Pub/Sub

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Aggregation levels



Project

A **project-level log sink** exports all the logs for a specific project.

A **log filter** can be specified in the sink definition to include/exclude certain log types.



Folder

A **folder-level log sink** aggregates logs on the folder level.

You can also include logs from children resources (subfolders, projects).



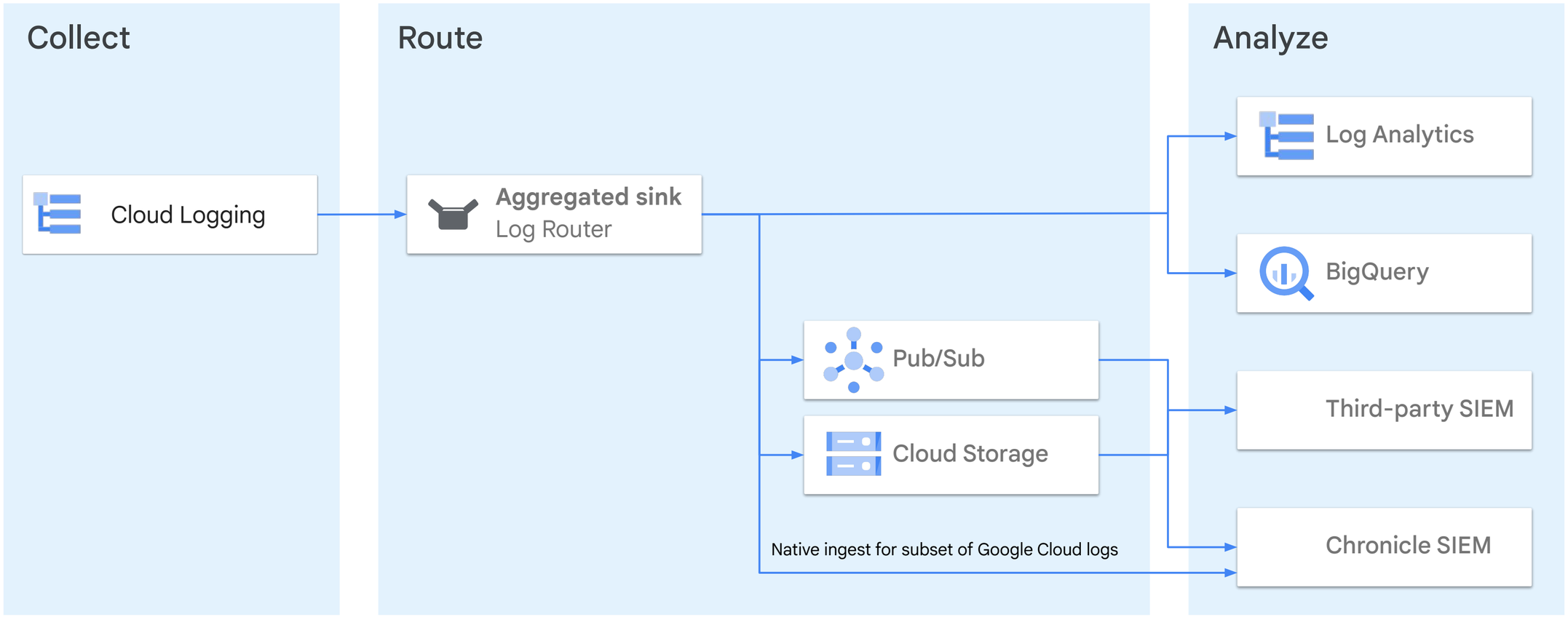
Organization

An **organization-level log sink** aggregates logs on the organization level.

You can also include logs from children resources (subfolders, projects).

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Security log analytics workflow recommends aggregated sinks



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Field naming

Log entry field	LogEntry type mapping	BigQuery field name
insertId	insertId	insertId
textPayload	textPayload	textPayload
httpRequest.status	httpRequest.status	httpRequest.status
httpRequest. requestMethod.GET	httpRequest. requestMethod.[ABC]	httpRequest. requestMethod.get
resource.labels.moduleid	resource.labels.[ABC]	resource.labels.moduleid
jsonPayload.MESSAGE	jsonPayload.[ABC]	jsonPayload.message
jsonPayload.myField. mySubfield	jsonPayload.[ABC].[XYZ]	jsonPayload.myfield. mysubfield

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Last three days from syslog and *apache_access* for a particular *gce_instance*

```
SELECT
  timestamp AS Time, logName as Log, textPayload AS Message
FROM
  (TABLE_DATE_RANGE(my_bq_dataset.syslog_,
    DATE_ADD(CURRENT_TIMESTAMP(), -2, 'DAY'), CURRENT_TIMESTAMP())),
  (TABLE_DATE_RANGE(my_bq_dataset.apache_access_,
    DATE_ADD(CURRENT_TIMESTAMP(), -2, 'DAY'), CURRENT_TIMESTAMP()))
WHERE
  resource.type == 'gce_instance'
  AND resource.labels.instance_id == '1554300700000000000'
ORDER BY time;
```

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Failed App Engine requests for the last month

```
SELECT
  timestamp AS Time,
  protoPayload.host AS Host,
  protoPayload.status AS Status,
  protoPayload.resource AS Path
FROM
  (TABLE_DATE_RANGE(my_bq_dataset.appengine_googleapis_com_request_log_,
    DATE_ADD(CURRENT_TIMESTAMP(), -1, 'MONTH'), CURRENT_TIMESTAMP()))
WHERE
  protoPayload.status != 200
ORDER BY time
```

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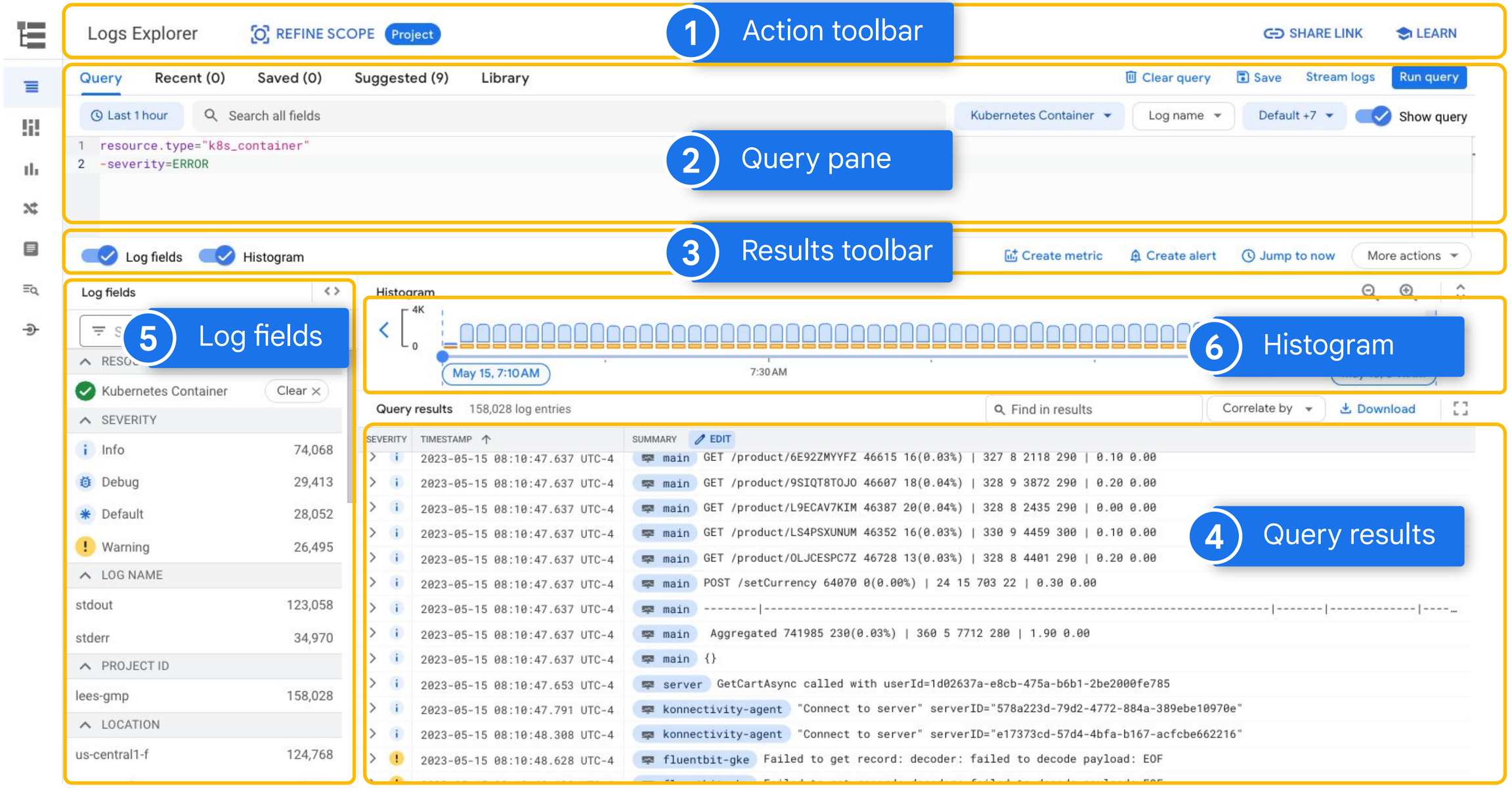
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Logs Explorer

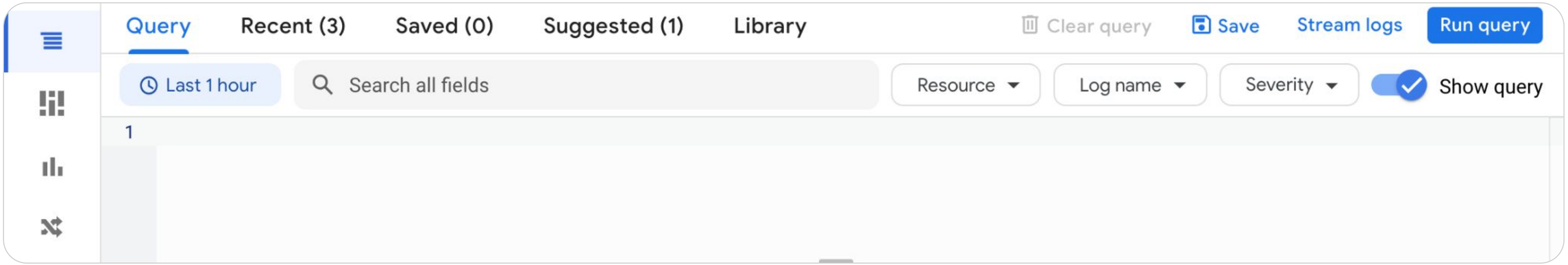
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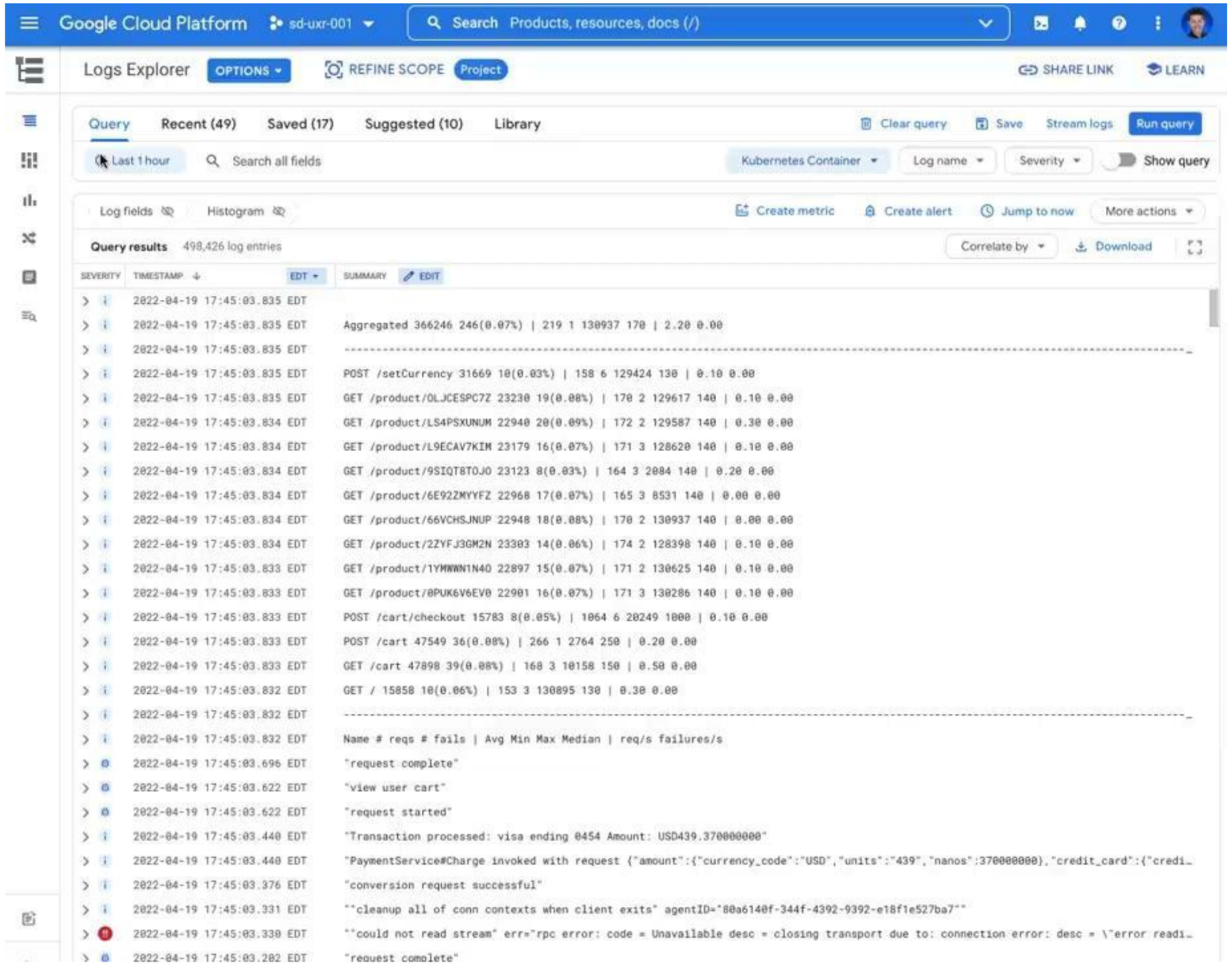
Ultimately, it’s the query that selects the entries

- Start with what you know about the entry you’re trying to find.
- If it belongs to a resource, a particular log name, or has a known severity, use the query builder drop-down menus.



Building queries made easier using dropdowns

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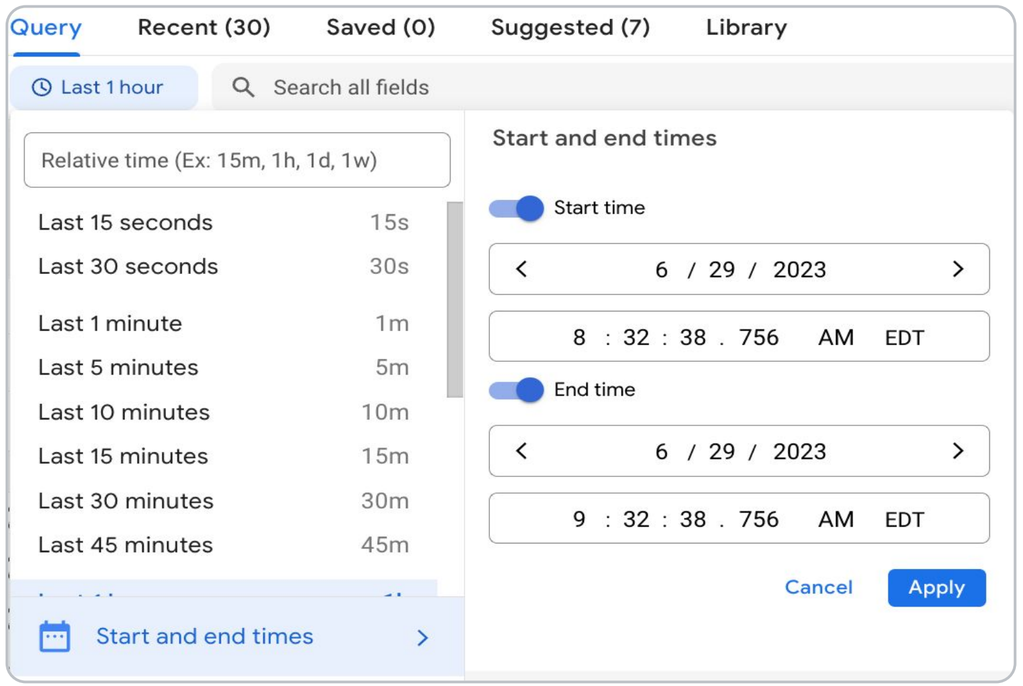
Comparison operators to filter queries

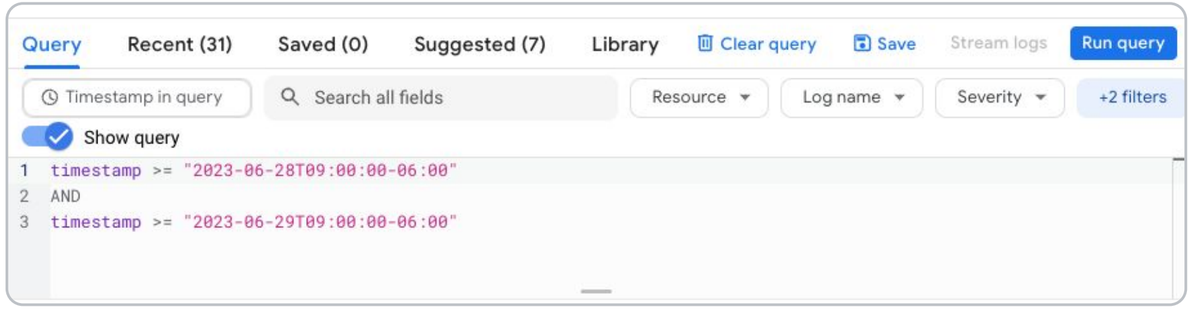
```
[FIELD_NAME] [OP] [VALUE]
```

=	Equals	`resource.type="gce_instance"`
!=	does not equal	`resource.labels.instance_id!="1234567890"`
> < >= <=	numeric ordering	`timestamp <= "2018-08-13T20:00:00Z"`
:	has	`textPayload:"GET /check"`
:*	presence	`jsonPayload.error:*`
=~	search for a pattern	`jsonPayload.message =~ "regular expression pattern"`
!~	search not for a pattern	`jsonPayload.message !~ "regular expression pattern"`

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Finding log entries, set the time range

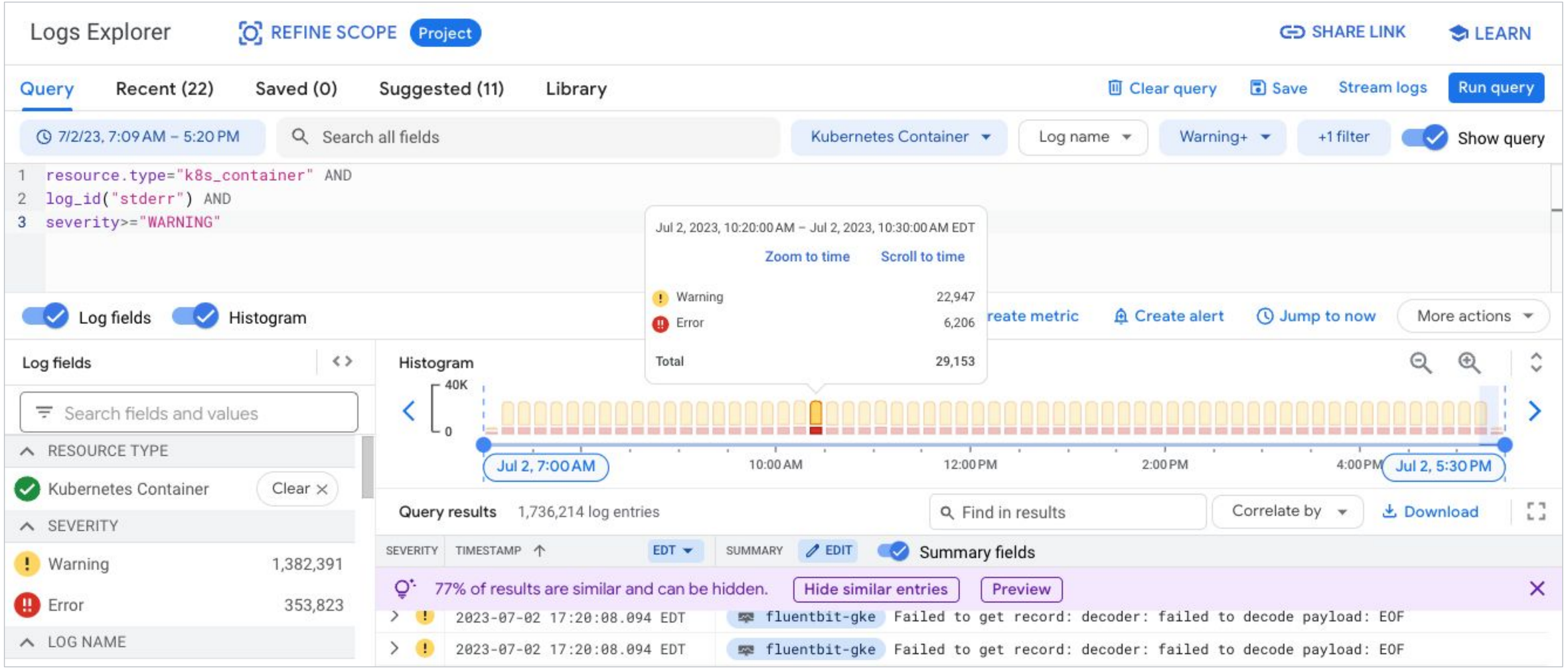




Narrowing the time range helps in quicker processing of queries.

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Narrow with Logs fields and Histogram



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Supported Boolean operators

AND	`textPayload:("foo" AND "bar")`
NOT	`textPayload:("foo" AND NOT "bar")`
OR	`textPayload:("foo" OR "bar")`

Ensure to use the all caps for the operator name.

The NOT operator has the highest precedence, followed by OR and AND in that order.

The Boolean operators AND and OR are short-circuit operators.

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The recipe for finding entries

- What do you know about the log entry?
  - Log filename, resource, a bit of text?
- Full text searches are slow, but may be effective:
  - **"/score called"**
- Use indexed SEARCH function for complete text matches, because they perform a case-insensitive match
  - SEARCH(textPayload, "hello world")
- If possible, restrict text searches to an log field
  - **jsonPayload:"/score called"**
  - **jsonPayload.message="/score called"**

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Finding entries quickly

- Search on an indexed field

```
httpRequest.status, logName, operation.id, resource.type, timestamp, severity
```

- Apply constraints on **`resource.type`** and **`resource.labels`** field

```
resource.type = "gke_cluster"
resource.labels.namespace = "my-cool-namespace"
```

- Be specific on which logs you're searching

```
logName="projects/benkelly-test/logs/apache-access"
```

- Limit the time range that you're searching

```
timestamp >= “2018-08-08T10:00:00Z” AND timestamp <= “2018-08-08T10:10:00Z”
```

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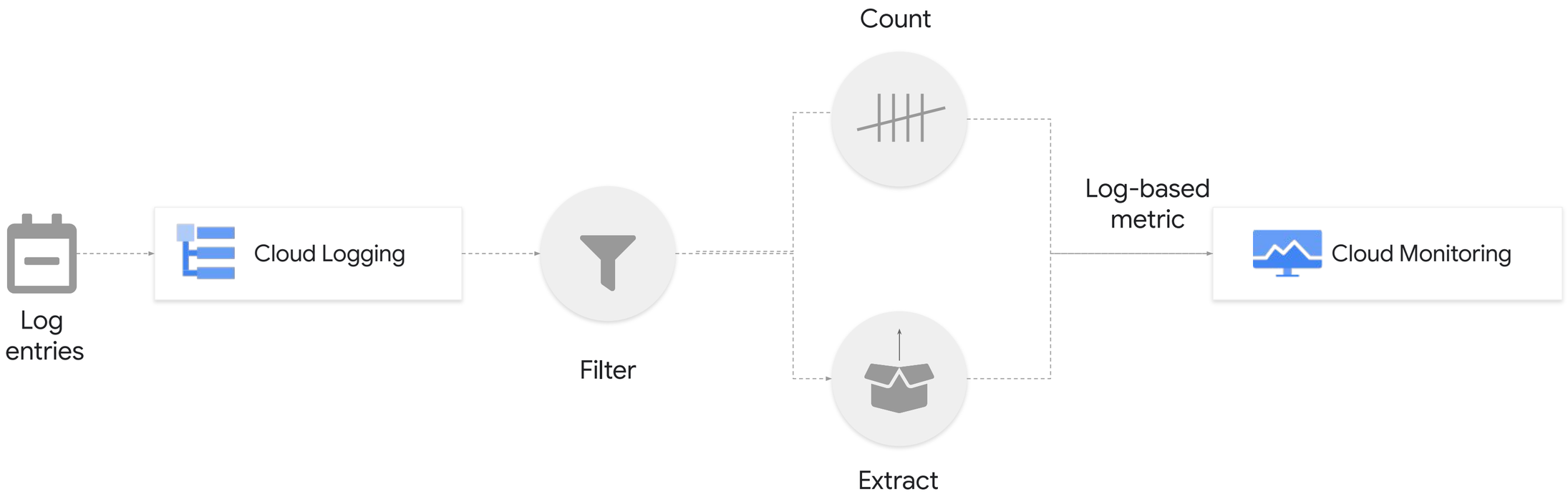
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Logs-based metrics



Logs-based metrics derive metric data from the content of log entries.

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Logs-based metrics are suitable in different cases

Count the occurrences

Count the occurrences of a message, like a warning or error, in your logs and receive a notification when the number of occurrences crosses a threshold.

Observe trends in your data

Observe trends in your data, like latency values in your logs, and receive a notification if the values change in an unacceptable way.

Visualize extracted data

Create charts to display the numeric data extracted from your logs.

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Key access control roles

- **Logs Configuration Writer**
  - List, create, get, update, and delete log-based metrics.
- **Logs Viewer**
  - View existing logs.
- **Monitoring Viewer**
  - Read the time series in log-based metrics.
- **Logging Admin, Editor, and Owner**
  - These are broad-level roles that can create log-based metrics.

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Log-based metric types

+1

Counter metrics

Count the number of matched log entries



Distribution metrics

Record the statistical distribution of the extracted log values

01

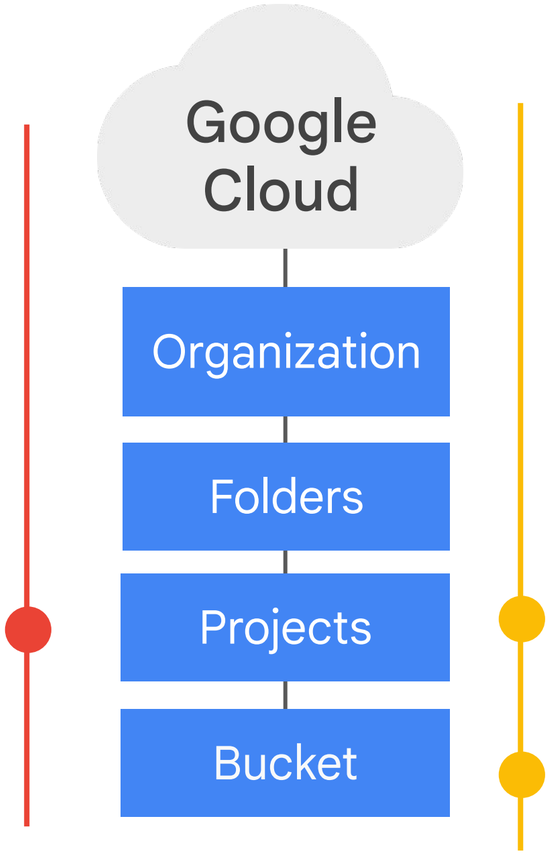
Boolean metrics

Record where a log entry matches a specified filter

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Scope of log-based metrics

System-defined log-based metrics apply at the project level



User-defined log-based metrics can apply either at project or bucket level

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A sample test code

```
//Basic NodeJS app built with the express server
app.get('/score', (req, res) => {
 //Random score, the contaierID is a UUID unique to each
 //runtime container (testing was done in Cloud Run).
 //funFactor is a random number 1-100
 let score = Math.floor(Math.random() * 100) + 1;
  console.log(`/score called, score:${score},
     containerID:${containerID}, funFactor:${funFactor}`);
   //Basic message back to browser
 res.send(`Your score is a ${score}. Happy?`);
 });
```

Track the request on the on the '/score' path

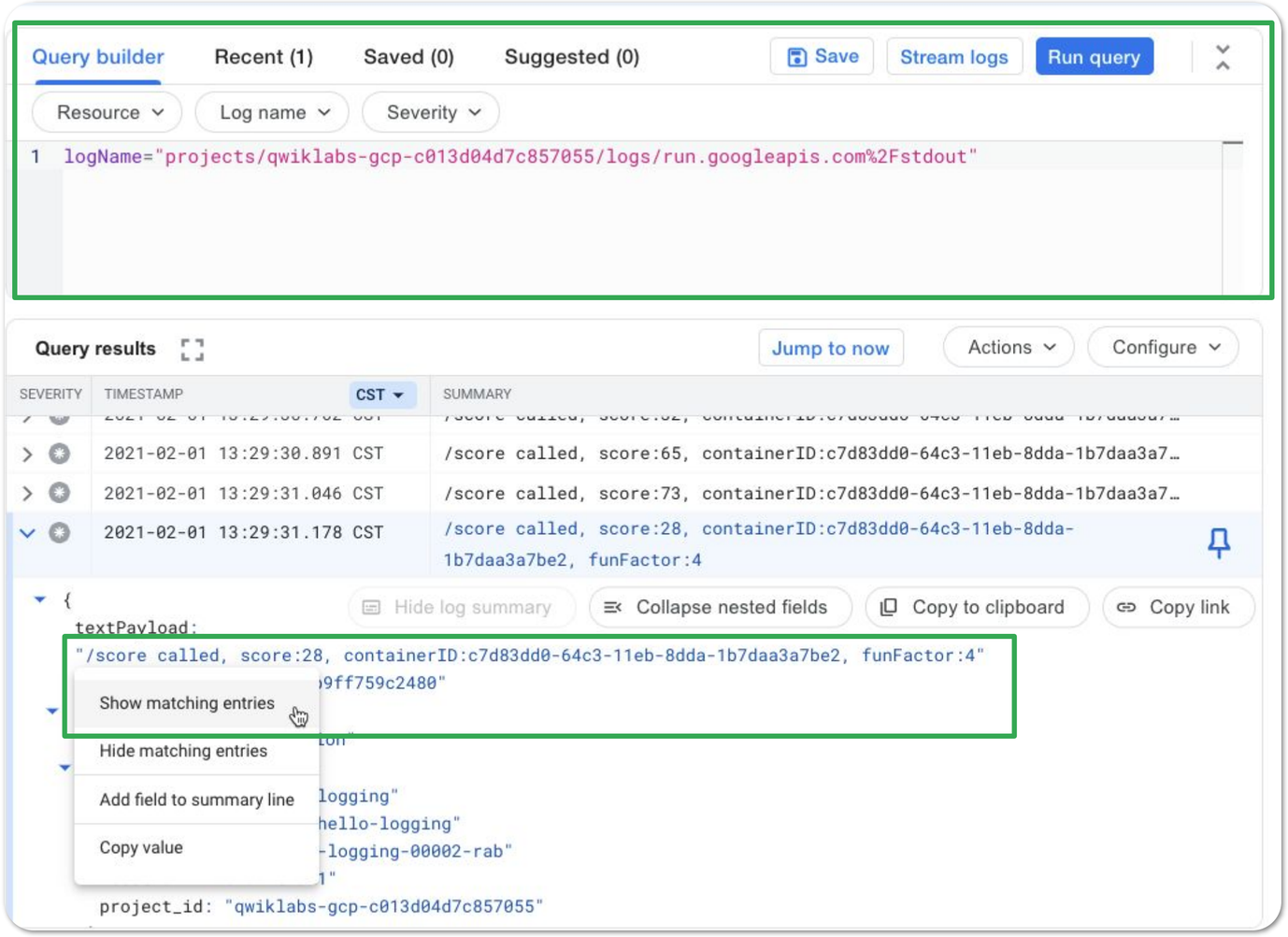
If a request matches, generate 1-100 score

Define log entry fields

Send a msg containing the score

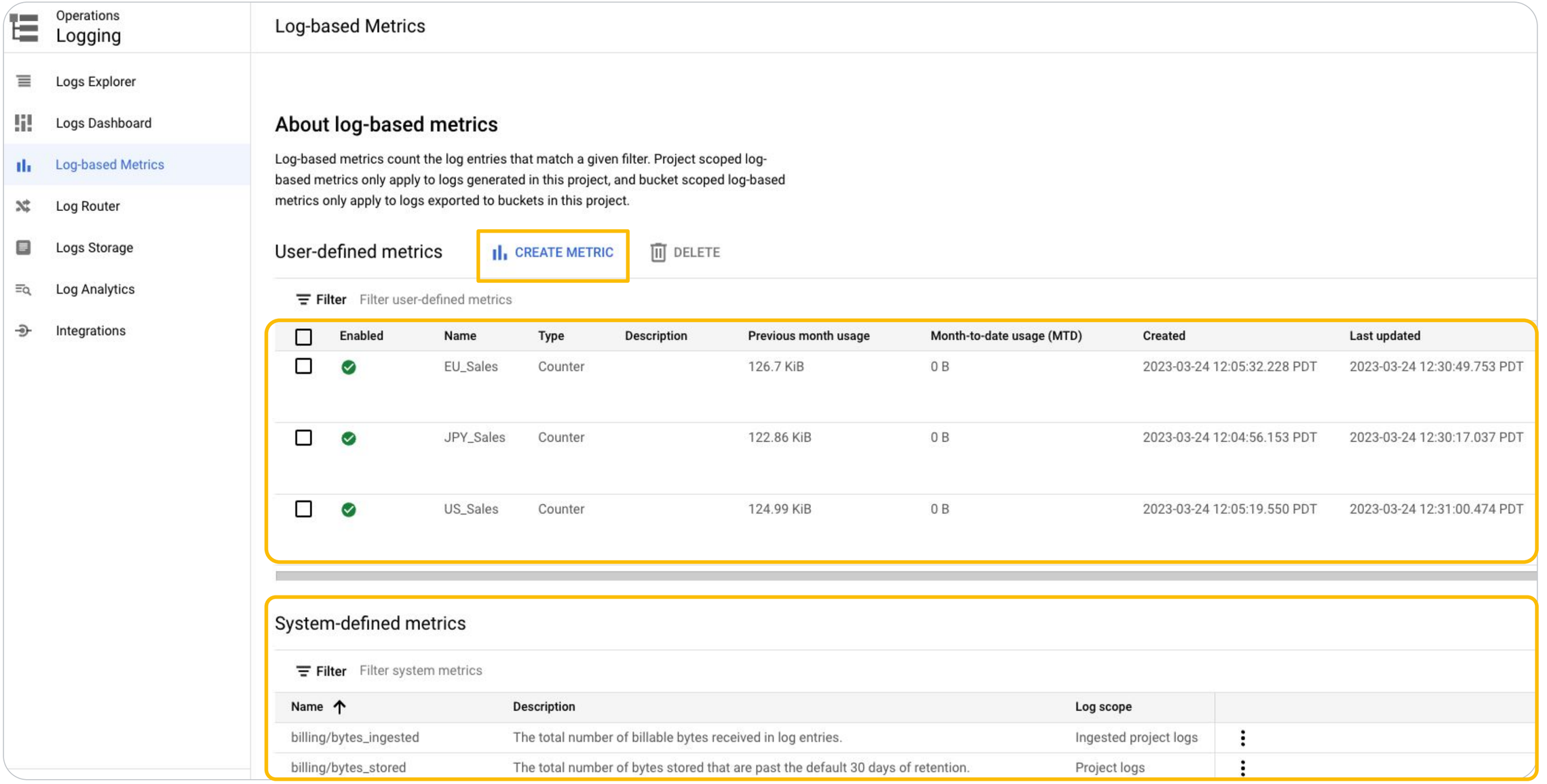
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Filtering entries



Log-based metrics

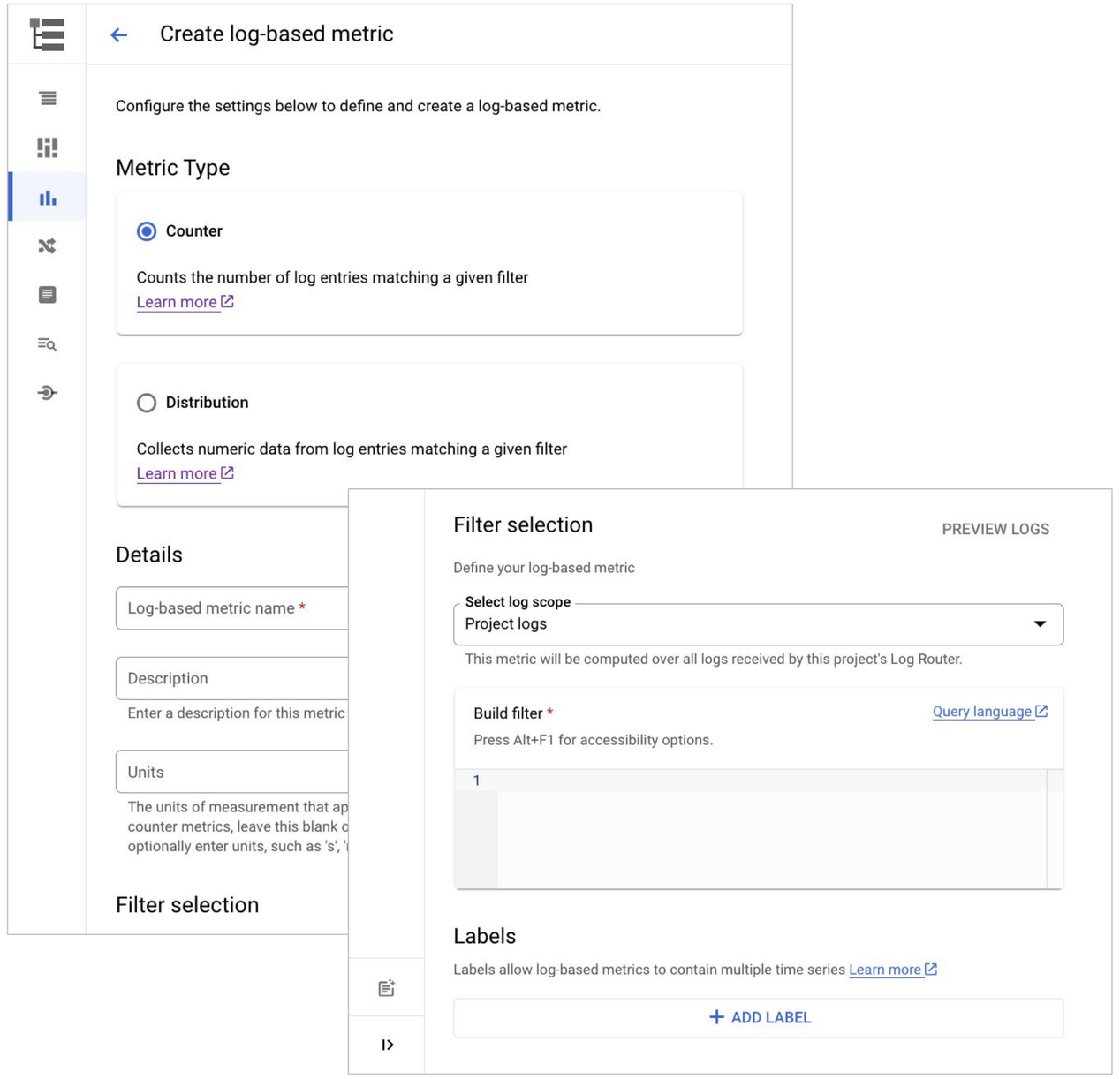
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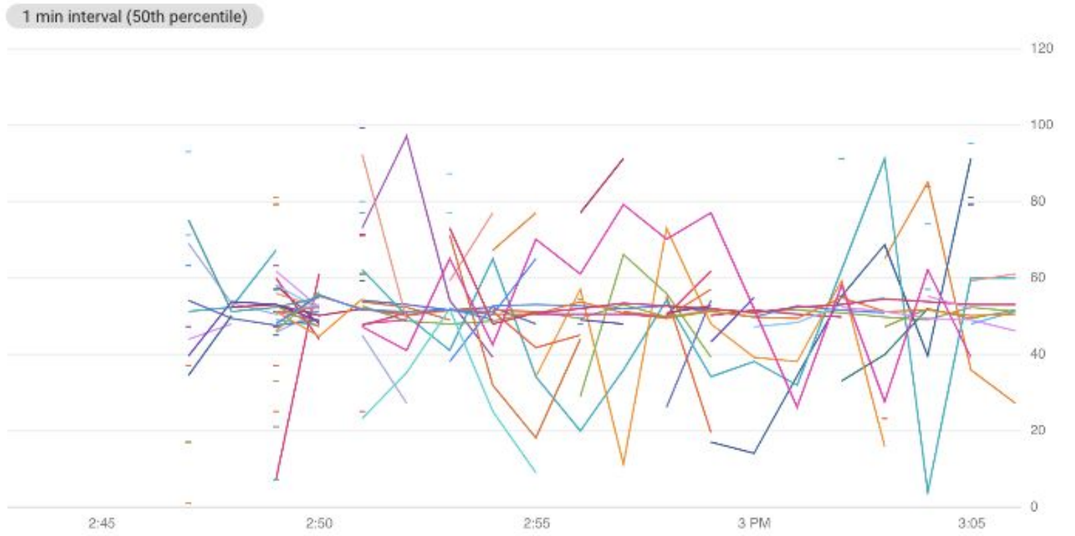
In Logs Explorer

1. Find the log with the requisite data
2. Filter to the required entries
3. **Actions | Create Metric**
4. Pick a metric type (Counter or Distribution)
5. If Distribution, set configurations
6. You can also add labels

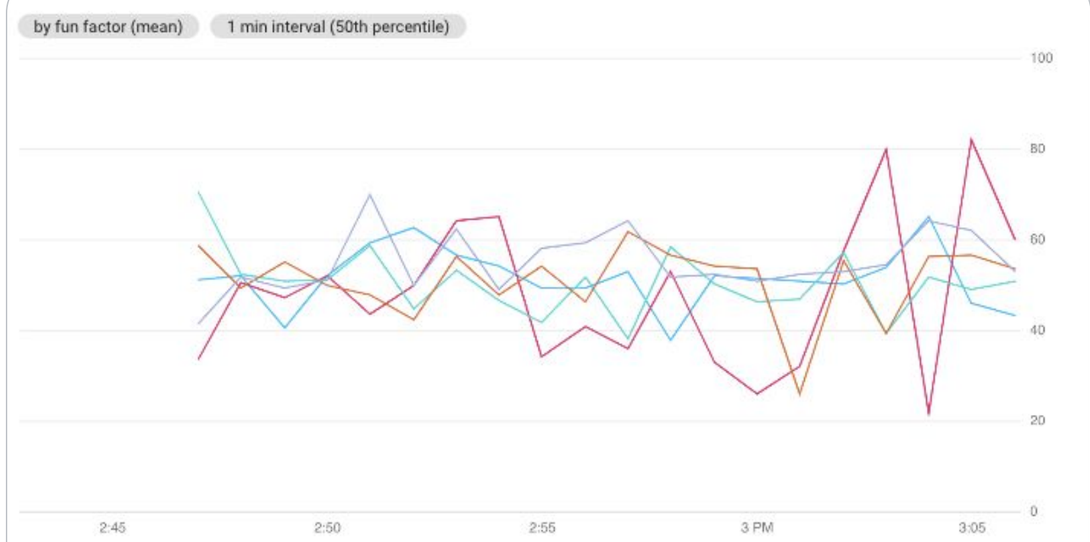


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Labels (for example, group-by, or filter)



container_instance	fun_factor	Value
036a1bb0-4923-11ea-b037-8f848c97d6cd	1	47.71
03eb1610-4924-11ea-9b7d-b75583332349	5	51.50
03efd100-4924-11ea-9ccf-551a2a7dd977	5	27.00
08698430-4922-11ea-b95f-3f69820c1eb5	2	52.33
0885e5d0-4922-11ea-b73e-d14a0894c3d9	1	47.00



fun_factor	Value
1	53.00
2	53.50
3	50.83
4	60.00
5	43.17

fun_factor

Google Cloud

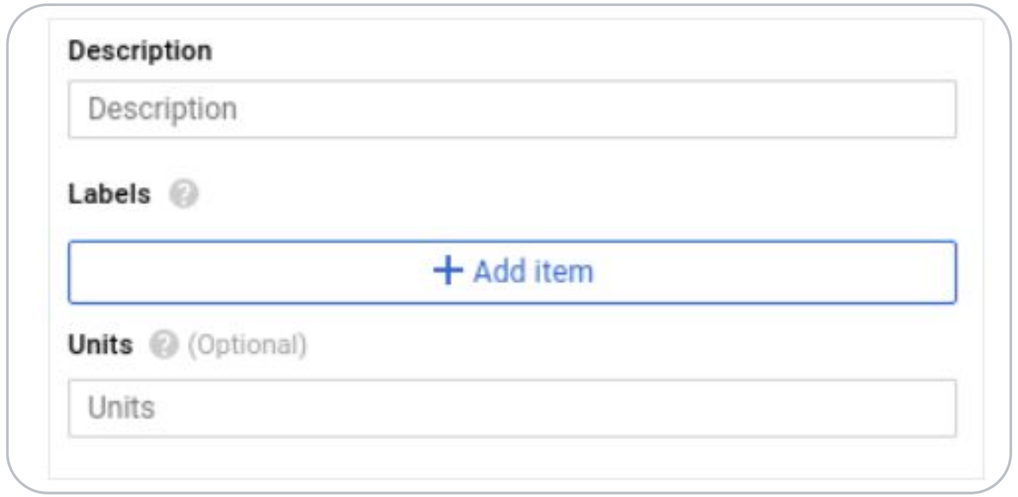
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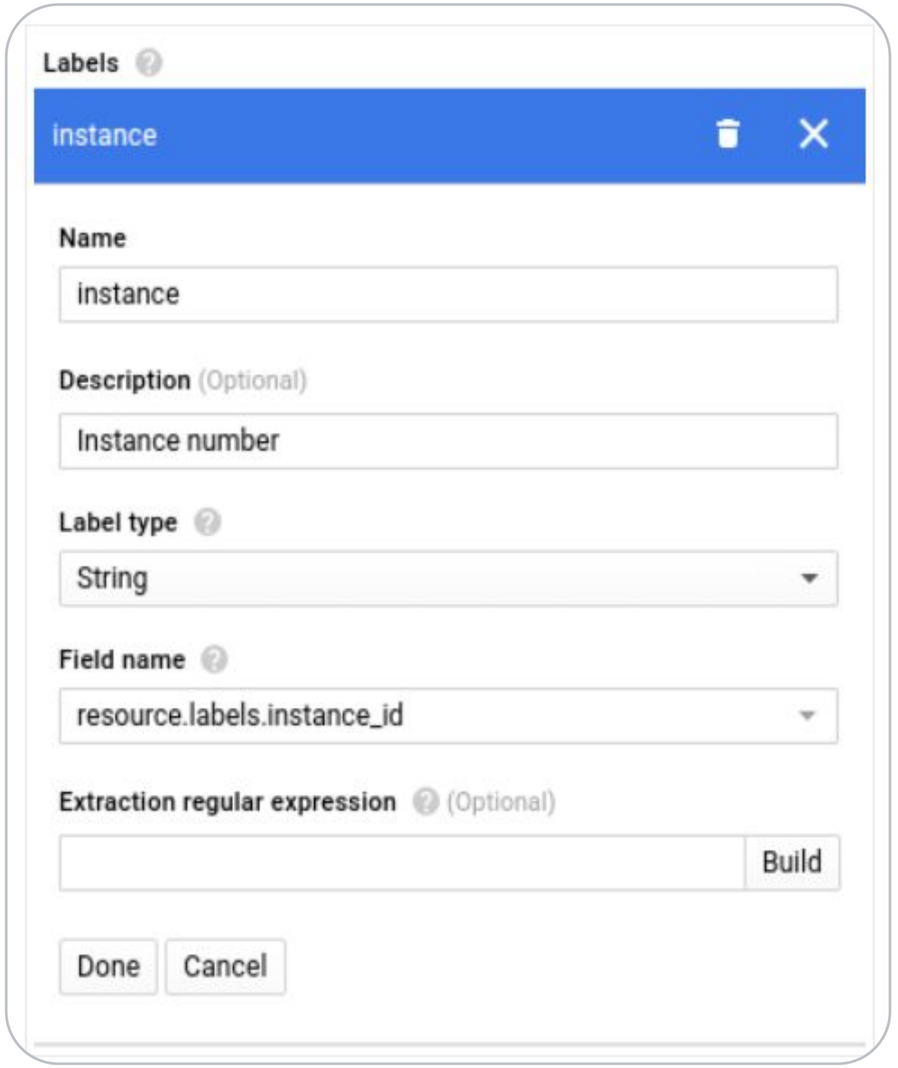
Labels and logs

- Labels allows for log-based metrics to contain a time series for each label.
- Two types of labels applied:
  - Default
  - User-defined
- User-defined labels can be either of the following:
  - The entire contents of a named field in the LogEntry object.
  - A part of a named field that matches a regular expression.
- You can create up to ten user-defined labels per metric.
- A label cannot be deleted once created.
  - Grows the time series significantly.

Creating user-defined labels

User-defined labels can be created when creating a log-based metric.





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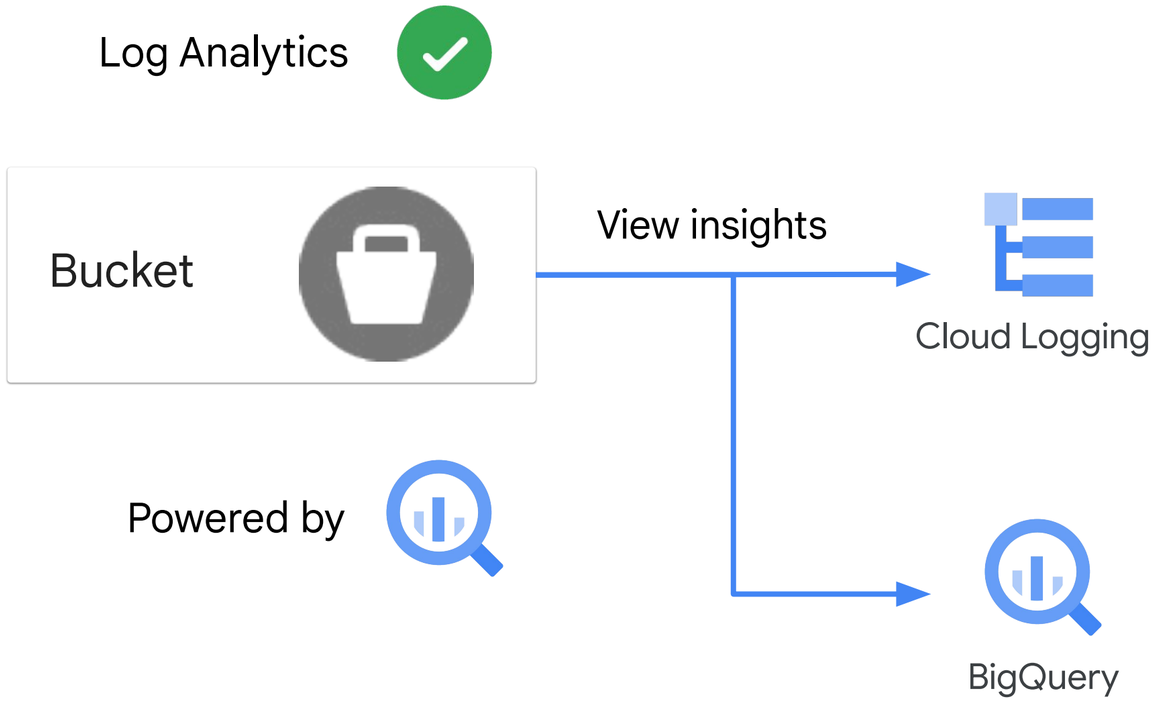
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- **Log Analytics**

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Perform analytics on log data using Log Analytics



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Cloud Logging and Log Analytics use cases

Troubleshooting

Logs Explorer



Get to the root cause with search, filtering, histogram and suggested searches.

Log Analysis

Log Analytics



Analyze application performance, data access and network access patterns.

Reporting

BigQuery link and Looker products



Use the same logs data in Log Analytics directly from BigQuery to report on aggregated application and business data found in logs.

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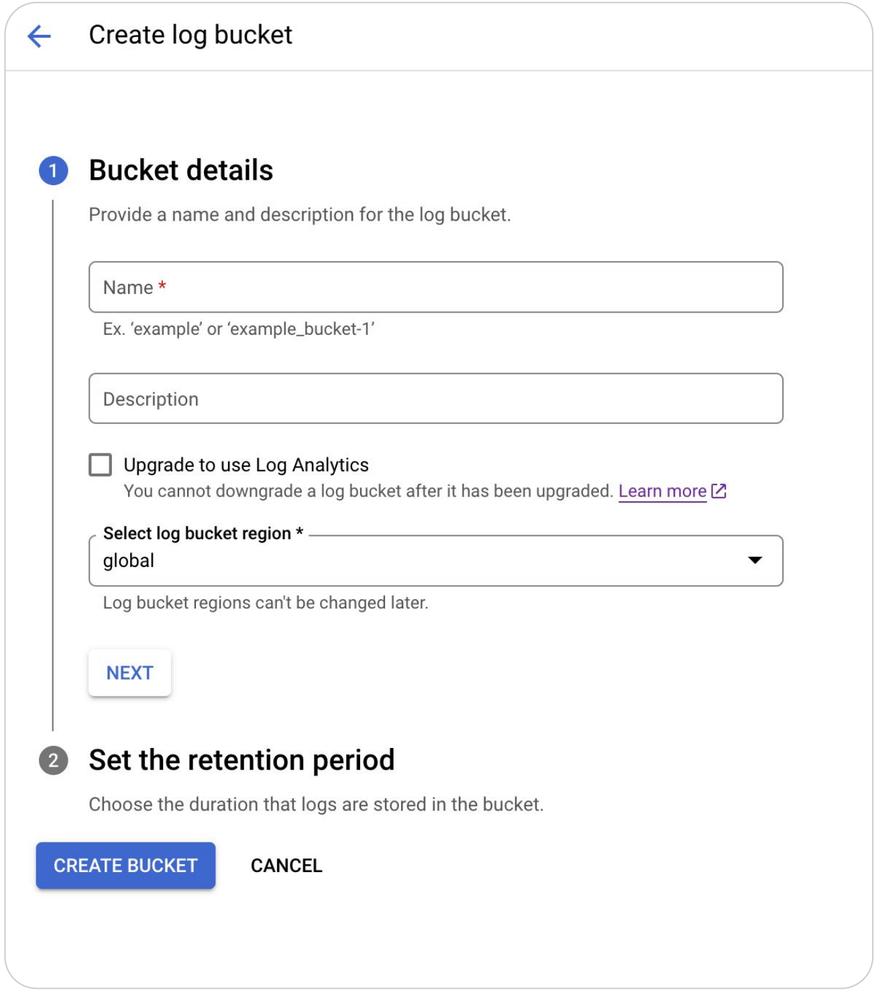
How different is analytics-enabled bucket log data from logs routed to BigQuery

- Log data in BigQuery is managed by Cloud Logging.
- BigQuery ingestion and storage costs are included in your Logging costs.
- Data residency and lifecycle are managed by Cloud Logging.

Creating a log-analytics enabled bucket

1. Create a log bucket with Log Analytics enabled.
2. Create a sink to route logs to the newly created bucket.
3. Check **Upgrade to use Log Analytics.**

You can't downgrade the log bucket to remove the use of Log Analytics.



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Log Analytics use cases

DevOps

Reduce MTTR by using advanced analytical capabilities to diagnose issues.

“Help me quickly troubleshoot an issue by looking at the top count of requests grouped by response type and severity”.

Security

Better investigate security-related attacks with queries over large volumes of security logs.

“Help me find the all the audit logs associated with a specific user over the past month”.

IT or Network Operations

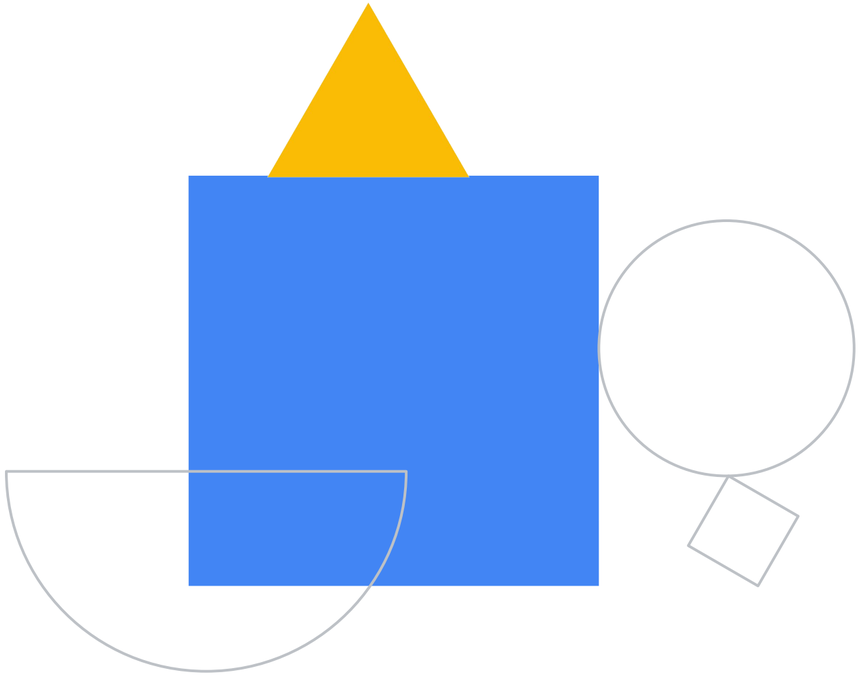
Provide Better network insight and management through advanced log aggregation capabilities

“Help me identify network issues for GKE instances using VPC and firewall rules”.

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Lab Intro

Log Analytics on Google Cloud



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Knowledge Check



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Quiz | Question 1

Question

You want to compare resource utilization for VMs used for production, development, and testing. What should you do?

A. Add a label called “state” to your VMs with the values “dev”, “test”, and “prod” and group by that label in your monitoring chart.

B. Put those resources in different projects and use Dataflow to create an aggregation of log values for each.

C. Name the VMs with a prefix like “dev-”, “test-”, and “prod-” and filter on the name property when reporting.

D. Export all machine logs to Cloud Storage and use Cloud Run functions to build reports based on the VM tags.

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Quiz | Question 1

Answer

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Quiz | Question 2

Question

Your manager wants a daily report of resource utilization by application. Where would the best export sink be?

A. Cloud Storage

B. Pub/Sub

C. BigQuery

D. Spanner

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Quiz | Question 2

Answer

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Recap

01 Use Log Explorer features

02 Explain the features and benefits of log-based metrics

03 Define log sinks (inclusion filters and exclusion filters)

04 Explain how BigQuery can be used to analyze logs

05 Use Log Analytics on Google Cloud



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Reference materials

- Log Analytics samples

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